

International Marketing Analytics

Logistic regression across countries

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Learning objectives

Choose the model: binary, multinomial, conditional, nested, ordinal or multilevel logit

Know the applications: entry mode, exporting, market selection, consumer choice, churn

Compare countries correctly: measurement invariance and the logit scale problem

Interpret results: odds ratios, average marginal effects, predicted probabilities

Test interactions: on the probability scale, not from the product term

Report to standard: a checklist for theses and papers

Session roadmap

1	2	3	4	5
Foundations	**Model family**	**Applications**	**Countries**	**Reporting**
Odds, probabilities, the S-curve	Who or what drives the choice	Entry, export, markets, consumers	Two layers of comparability	Marginal effects, checklist, data

One rule for today: compare probabilities across countries, not coefficients.

Before we start: does R&D matter more in Germany?

Two logit models of export status, one per country:

R&D coefficient: 0.8 in Germany, 0.5 in Hungary.

What does 0.8 mean?
Log-odds, odds or probability?

Can you compare 0.8 and 0.5?
What else differs between the two samples?

What would you report instead?
Think about what a manager wants to know.

Foundations: from odds to probabilities

Part 1 of 5 · why logit, three scales, odds ratios

Why not ordinary regression for a yes/no outcome?

Linear probability model

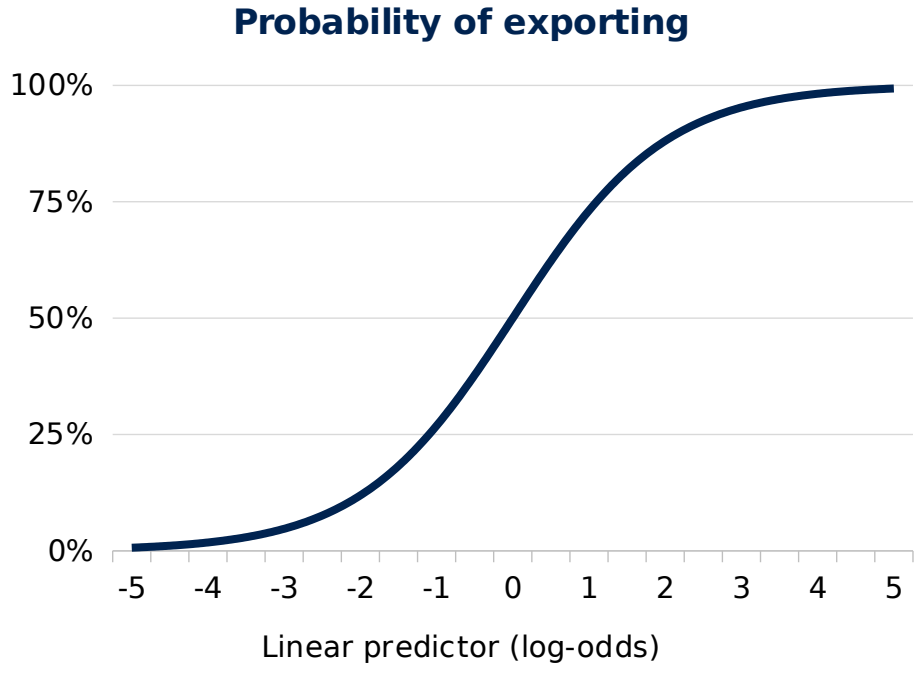
- Predictions can fall below 0 or above 1
- Same effect at every baseline
- Useful as a robustness check

Logistic regression

- Probabilities stay between 0 and 1
- Largest effect near 50 %, smallest at the extremes
- Estimated by maximum likelihood

The price of the S-curve: a coefficient no longer tells you how much the probability changes.

One model, three scales



Log-odds: β adds up linearly. This is what the output table shows.

Odds: e^β multiplies the odds. This is the odds ratio.

Probability: what managers ask about. Its change depends on the starting point.

Worked example: from counts to odds

German SMEs	Exporters	Firms	Probability p	Odds = p / (1 − p)
With R&D	80	100	0.80	0.80 / 0.20 = 4.0
Without R&D	60	100	0.60	0.60 / 0.40 = 1.5

Formula

```
p    = exporters / firms
odds = p / (1 − p)
     = exporters / non-exporters
```

Odds 4.0: four exporters per non-exporter

Odds 1.5: three exporters per two non-exporters

Scales: probability from 0 to 1, odds from 0 to infinity; odds of 1 mean 50:50.

Odds ratio and log-odds: the same example

2.67 **Odds ratio:** 4.0 / 1.5; R&D multiplies the odds by 2.67

1.39 **Log-odds with R&D:** ln 4.0; without R&D ln 1.5 = 0.41

0.98 **Difference = ln 2.67:** the logit coefficient of R&D

Back to probability

```
odds → p:
 p = odds / (1 + odds)
   = 4.0 / 5.0 = 0.80

log-odds → p:
 p = 1 / (1 + e^−1.39)
   = 0.80
```

Not “2.67 times as likely”: the probability rises from 0.60 to 0.80, by 20 pp or a factor of 1.33.

Your turn: Hungarian SMEs, with a calculator

Probability → Odds → Odds ratio → Log-odds → Back to p

Hungarian SMEs	Exporters	Firms	p	Odds	Log-odds
With R&D	60	100	?	?	?
Without R&D	45	100	?	?	?

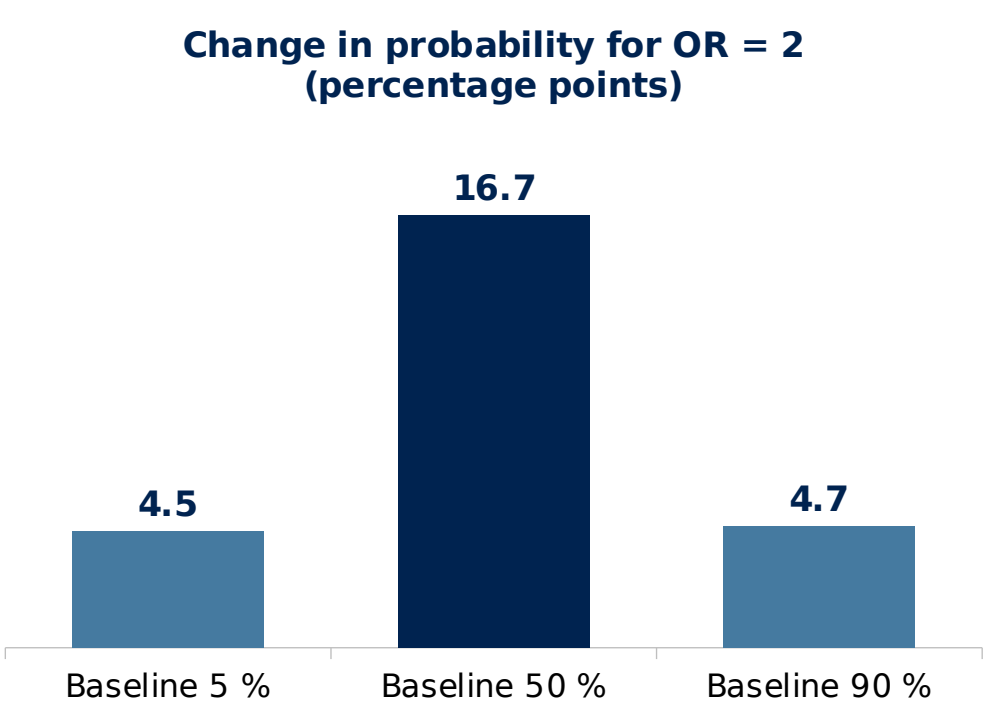
1–2 probability and odds of both groups; the odds ratio

3–4 log-odds and their difference; from log-odds 0.41 back to p

5 compare with Germany: odds ratio and the gap in percentage points

Python check: `smf.logit("exporter ~ rd", data=df).fit().params`

An odds ratio of 2 is not “twice as likely”



Calculation: $p_1 = OR \cdot p_0 / (1 - p_0 + OR \cdot p_0)$

Read the chart

Doubling the odds moves

5 % to 9.5 %

50 % to 66.7 %

90 % to 94.7 %

“Twice as likely” only holds for rare outcomes, below about 10 %.

Choosing the model

Part 2 of 5 · who or what drives the choice

Does the chooser or the option drive the choice?

Chooser: multinomial logit

- Variables describe the firm or consumer: size, experience, home culture
- Wide data: one row per decision
- Example: entry mode by asset specificity

Options: conditional logit

- Variables describe the alternatives: host region, product profile
- Long data: one row per decision × alternative
- Example: plant location by agglomeration

Ask first: does my theory explain the chooser or the options?

Greene, Chidlow & Strange (2022), International Business Review.

The logit family at a glance

Outcome	Model	Assumption to defend	IM/IB example
Two outcomes	Binary logit / probit	Link is mostly convention	Anderson & Coughlan 1987
3+ unordered, chooser traits	Multinomial logit	IIA	Kogut & Singh 1988
Options with attributes	Conditional logit	IIA, correct choice set	Head, Ries & Swenson 1995
Staged decisions	Nested / sequential logit	Nests match the decision	Pan & Tse 2000
Taste heterogeneity	Mixed logit	Distribution of random coefficients	Rasciute & Pentecost 2014
Ranked outcomes	Ordered logit	Proportional odds (Brant test)	Few IM examples
Firms in countries	Multilevel logit	Enough countries	Arrégle et al. 2006
Rare events	Firth / rare-events logit	Small-sample bias, separation	Jiménez et al. 2019

IIA: the assumption behind multinomial logit

What IIA demands

- The odds between two modes must not change when a third is added
- Implausible when greenfield and acquisition are close substitutes
- Standard IIA tests contradict each other (Cheng & Long 2007)

A staged decision

1. Export or FDI?
2. Greenfield or acquisition?
3. Wholly owned or joint venture?

Defend MNL with theory about substitutes, then check with a nested or mixed logit.

Firms nested in countries: how many countries?

25–30 countries as a rough minimum for country-level inference (Bryan & Jenkins 2016)

7 countries in EU-EFIGE: use country fixed effects, not country random effects

120+ economies in Global Findex: enough for country-level moderators

Cross-level interaction? Add a random slope, e.g. for firm experience when testing uncertainty avoidance × experience.

Applications in international marketing

Part 3 of 5 · entry, export, markets, consumers

Entry mode: four decades, four model generations

Binary	Multinomial	Nested	Multilevel
1987	**1988**	**2000**	**2006–25**
Anderson & Coughlan: integrated vs independent channels	Kogut & Singh: 228 US entries, cultural distance	Pan & Tse: 10,000+ entries in China, equity choice first	Arrégle et al.; Mas-Ruiz et al.: 3,023 FDIs, 100 countries

Each step fixed a weakness of the one before: too few options, IIA, then data nested in firms and countries.

Meta-analyses: why cultural distance looks inconsistent

38 studies, 106 effect sizes: TCE effects as predicted, but moderated (Zhao, Luo & Suh 2004)

66 samples: no overall link between cultural distance and entry mode (Tihanyi et al. 2005)

230 studies with 1,499 effect sizes in a meta-analytic SEM (Wan et al. 2023)

1,217 tests in 119 studies: interaction terms often misread (Wulff)

Inference, not proof: part of the mixed record may be methodological.

Exporting: sunk costs make firms keep exporting

60 pp — higher export probability after prior exporting, Colombian plants (Roberts & Tybout 1997)

≈50 % — higher chance of exporting tomorrow when exporting today (Bernard & Wagner)

≤40 % — of that persistence is sunk cost once learning is modelled (Timoshenko)

Teaching point: logit or probit matters little; the dynamic structure matters a lot.

Market selection: conditional logit on long data

Plant	Option	Chosen	Cluster
1	Ohio	1	high
1	Texas	0	medium
1	Kentucky	0	low
2	Ohio	0	high
2	Texas	1	medium
2	Kentucky	0	low

Illustration of the layout, not real data

Head, Ries & Swenson (1995)

- 751 Japanese plants in US states
- Industry and keiretsu clusters attract new plants

Later refinements

- Mixed logit for taste heterogeneity
- Country first, region second
- Do not pool greenfield and M&A

Consumers: choice experiments across countries

Country of origin

- Conjoint with about 1,000 consumers per country in five countries
- Multi-cue designs show smaller origin effects
- “Made in Germany” fades once brand attitude is controlled

Cross-national segments

- Mixture segments across 11 EU countries (1999)
- “Countries as segments” fitted worst (2002)
- Five segments cut across borders

Segments cross borders: the country is not the default unit of analysis.

Adoption and diffusion: culture or wealth?

≈6 years to takeoff in 16 European countries, almost half as long in Scandinavia as in the Mediterranean

3,283 consumers in 11 countries: innovativeness with cross-level effects of national culture

≈75 % of explained variance in mobile-money adoption across 36 African countries is individual

Culture or wealth? 16 to 31 countries give unstable answers at the country level.

Churn and CRM: logit stays a strong benchmark

Tournament

- Neslin et al. 2006
- Logit and trees among the most accurate
- Method gaps worth real money

Benchmark

- Verbeke et al. 2012
- 11 telecom datasets worldwide
- Many classifiers perform alike

Profit

- ProfLogit, 2018
- Logit tuned for expected profit
- Best profit across 9 datasets

Open question for multinational CRM: does a churn model built in one country work in another? No study has tested it.

Comparing countries

Part 4 of 5 · two layers of comparability

Two comparability problems stack up

Layer 1: measurement

- Do survey items mean the same in each country?
- Response styles shift the answers
- Fix: invariance tests, IRT, method factors

Layer 2: logit scale

- Coefficients are scaled by unexplained variation
- That variation differs by country
- Holds even with perfect measurement

You need both: invariance tests fix layer 1; probabilities and marginal effects fix layer 2.

Layer 1: response styles differ by country

28 % of 243 cross-national marketing articles tested measurement invariance (He, Merz & Alden 2008)

26 countries: acquiescence and extreme answers track power distance and uncertainty avoidance (Harzing 2006)

15 countries: yea-saying highest in Brazil and India, nay-saying in the Netherlands and Japan

Configural, metric, scalar: test invariance before comparing relationships or means (Steenkamp & Baumgartner 1998).

Layer 2: same effect, different coefficients

Country	True effect	Noise scale	Logit coefficient
Germany	0.8	1.0	**0.80**
Hungary	0.8	1.6	**0.50**

Illustrative numbers

Latent-variable view

```
coefficient =
  true effect / scale

0.8 / 1.6 = 0.50
```

Compare probabilities: AMEs and predictions by country

Model the variance: heterogeneous-choice logit

Nested models: KHB decomposition

The gap in the warm-up can be noise, not strategy.

In choice experiments: compare scale-free quantities

1993 Swait & Louviere: test scale equality before you compare part-worths across samples

10/10 datasets preferred models with scale heterogeneity over standard mixed logit (Fiebig et al. 2010)

7 % of 278 group-comparing choice experiments formally accounted for scale (Wright et al. 2018)

Compare across countries: willingness-to-pay ratios, relative importances and simulated choice shares, not part-worths.

Interpreting and reporting

Part 5 of 5 · marginal effects, interactions, checklist

Three quantities to report

Odds ratio

What: multiplies the odds

Use: within one model

Trap: misleads for common outcomes

Marginal effect

What: average change in probability, in percentage points

Use: across models and countries

Prediction

What: probability for a profile, e.g. a typical Hungarian SME

Use: talking to managers

Standard: marginal effects and predictions with confidence intervals; odds ratios stay in the table.

Interactions: the product term is not the effect

Ai & Norton (2003)

- The effect on probability varies by firm
- It can be non-zero when the coefficient is zero
- It can even have the opposite sign

Three steps (Mize 2019)

1. **Plot** predicted probabilities across the moderator
2. **Test** marginal effects at chosen moderator values
3. **Compare** second differences between those effects

Managers ask about probabilities: “does distance matter more for SMEs?”

In Python: marginal effects with statsmodels

Python: statsmodels

```
import numpy as np
import statsmodels.formula.api as smf

m = smf.logit(
    "exporter ~ rd + np.log(employees)"
    " + C(country, Treatment('DE'))",
    data=df).fit()

m.get_margeff(dummy=True).summary()
ame_rd(m, df)                  # by country
m.get_prediction(profiles).summary_frame()

(ggplot(grid, aes("employees", "predicted",
                  color="group")) + geom_line())
```

`get_margeff` average marginal effects, in probability units

`ame_rd` effect of R&D by country: predict with rd = 1 vs 0

`get_prediction` probability with 95 % CI for a firm profile

`plotnine` the curve your reader will understand

Full code and data: sessions/01-foundations/logit_exports.qmd, simulated EFIGE-style SMEs.

Reporting checklist for theses and papers

Element	Report at least	Avoid
Focal effects	Marginal effects with SEs; predictions for profiles	“x times as likely” for common outcomes
Moderation	Plot across the moderator with CIs; second differences	The product-term p-value alone
Countries	Marginal effects by country; heterogeneous choice; WTP ratios	Betas or ORs across country models
Mediation	KHB decomposition	Coefficients across nested logits
Fit	LR test, one labelled pseudo-R^2, AUC	% correct at a 0.5 cut-off with rare events
Multilevel	Number of countries; random slopes	Culture effects from < 25 countries as proof
Endogeneity	Argued exclusion restriction; control function	Fitted values plugged into a logit

Endogeneity: IV logic does not carry over to logit

54 of 165 strategy papers claimed a valid exclusion restriction for Heckman (Wolfolds & Siegel 2019)

83 % of 246 top-IB studies met only half of a minimum standard for controls (Nielsen & Raswant 2018)

Do: control functions or two-stage residual inclusion, with an argued instrument.

Don't: plug first-stage fitted values into a logit, or run Heckman without an instrument.

Open data for international logit exercises

Dataset	Coverage	Binary outcome	Access
EU-EFIGE	~15,000 firms, 7 EU countries, 2007–09	Exporter yes/no	Open, CC-BY 4.0
Flash Eurobarometer 421	SMEs in the EU, 2015	Exports inside/outside EU	GESIS ZA6585
Flash Eurobarometer 397	Consumers, EU countries, 2014	Bought online cross-border	GESIS ZA5943
World Bank Enterprise Surveys	270,000+ firms, 177 economies	Direct or indirect exporter	Free, registration
Global Findex 2021	~128,000 adults, 120+ economies	Digital payment	Open, World Bank
UCI Bank Marketing	41,188 contacts, Portugal	Term deposit (~11 % yes)	Open

A six-session teaching sequence

Session	Concept	Exercise
1	Why not OLS; odds vs probability	One OR at baselines of 5, 50 and 90 %
2	Marginal effects, predictions, fit	Bank Marketing lift chart; QWE churn case
3	Entry mode: binary, MNL, nested; IIA	Mode probabilities by distance
4	Interactions in nonlinear models	FL397: trust × country on the probability scale
5	Comparability: scale, invariance, multilevel	EFIGE: German vs Hungarian SME
6	Choice experiments and location choice	Two-country conjoint: compare WTP ratios

Quiz: odds, countries and interactions

1 OR = 2, baseline 50 %. New probability?
A 67 % B 100 % C 52 %

2 R&D coefficient 0.8 in Germany, 0.5 in Hungary. Conclusion?
A larger effect in Germany B none yet C negative in Hungary

3 The interaction coefficient is not significant. The effect on probability…
A is negative B is zero C can still be non-zero

4 Plants choose among US states described by their attributes. Model?
A conditional logit B multinomial logit C ordered logit

5 Firms from 7 countries, Hofstede moderators in a multilevel logit. Results are…
A confirmatory B unbiased C exploratory at best

3 minutes alone, then compare with your neighbour.

Your turn: export probability, Germany vs Hungary

Fit · 10′ → **AMEs · 10′** → **Compare · 10′** → **Report · 5′**

1	**Fit** a logit of exporter status on R&D, size and country
2	**Compute** the average marginal effect of R&D, overall and by country
3	**Compare** coefficients from separate country models with the marginal effects
4	**Report** the export probability of a typical German and Hungarian SME, with CIs

Deliverable: one slide per pair with the plot and two sentences for a manager.

Open niches for theses

1	**Ordered logit** on degree of control or degree of marketing adaptation
2	**Conditional logit** for retailer and franchisor country selection
3	**Churn models** tested for transfer from one country to another
4	**Multilevel logit** of consumer AI adoption across 25 or more countries
5	**Joint modelling** of response styles and logit scale

Before you commit: run your own literature search; these are gaps in our search, not proof of absence.

Key takeaways

1	**Theory picks the model:** chooser traits → MNL, option attributes → conditional logit
2	**Odds are not probabilities:** an odds ratio's meaning depends on the baseline
3	**Two layers:** measurement invariance and the logit scale both threaten country comparisons
4	**Report probabilities:** marginal effects and predictions with confidence intervals
5	**Interactions on the right scale:** plot, test, compare; never the product term alone

Next: run sessions/01-foundations/logit_exports.qmd before class; it reproduces the warm-up.

Key sources and further reading

Methods

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Thank you. Questions and discussion

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